

Work and Energy¹

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This article was prepared by the author to serve as a draft for TNSCERT class 7 chapter on the concept of Work and Energy during the the Tamil Nadu Curriculum Framing (TNCF) workshop that was held in Asha Nivas, Chennai from 13-11-2025 to 15-11-2025, 09-03-2026 to 10-03-2026, and 27-04-2026 to 30-04-2026.

This draft assumes that the student knows: Force effect on motion, heat (thermal energy),

Cant assume understanding of difference between distance and displacement, acceleration due to gravity,

Draft Syllabus:

C 1.11: Understands and calculates work and energy, identifies different forms and sources of energy, observes and represents energy changes in real-life situations.

Work and Energy: Work: Concept and definition of work in science - quantification of work done by a steady force, Simple numerical problems, SI Unit of Work (Joule)

C 1.12: Applies scientific knowledge and uses appropriate scientific vocabulary to communicate ideas and to promote responsible use of energy.

Energy: Concept and definition of energy, different forms of energy, conversion of energy, energy flow in daily life systems, useful and wasted forms of energy, law of conservation of energy, sources of energy - renewable and non-renewable energy sources-conservation

In earlier lessons, we learned that a force is needed to start motion, stop motion, change speed, or change the direction of motion. Let us do a simple activity. Place a book on the desk and push it slowly so that it slides at a steady speed. By pushing slowly, we are trying to apply a constant force to move the book. In everyday language, we say that we are doing some *work*. If we push the same book a longer distance or if we are pushing a heavier stack of books the same distance, we are doing more work than we did initially.

Illustration of a hand sliding a book across a table.

Illustration of a hand sliding a book across a table for a long distance.

Illustration of a hand sliding a stack of books across a table.

From this, we can see that the work done depends on two things:

Figure 1: Sliding a book across a table.

1. how much force is applied, and
2. how far the object moves.

When a constant force (F) moves an object through some distance (d) in the same direction as the force, the work done (W) is defined as:

$$W = F \times d \quad (1)$$

The SI unit of work is Joule (J), named for James Prescott Joule, an English scientist of the 1800s.

The SI unit of force is Newton (N) and the SI unit of distance is metre (m). This makes

$$1 \text{ J} = 1 \text{ N} \times 1 \text{ m} \quad (2)$$

EXAMPLE: If we apply a force of 10 N to push the book a distance of 2 m, the work done = $10 \text{ N} \times 2 \text{ m} = 20 \text{ J}$.

QUESTION: In which of the following cases is work being done? In which of these cases are we applying force but there is no movement?

1. Holding a heavy bag
2. Pushing a wall
3. Climbing stairs
4. Kicking a ball
5. Sitting and reading

Remember that we are not concerned here with the common meaning of the word “work,” which implies that any physical or mental labor is work. We are only concerned with work where there is a force acting on an object and the object moves in the direction of the force.

QUESTION: Why is pushing a wall not considered work in physics?

When is work done zero?

From the definition of work done (Equation 1), we can see that work done is zero if no force is applied by the agent, i.e., $F = 0$. If the object does not move, $d = 0$, then no work done even if you are applying a force. For example, if you push a wall and it does not move, the work done is zero.

Work is done only when the force acts in the same direction as the movement. If the force acts in a direction perpendicular (at a right angle) to the movement, then the work done by that force is zero.

EXAMPLE: Thulasi pushes a book across the table, and the book moves forward. At the same time, Vaishali presses the book downward into the table. The book does not move downward. Thulasi is doing work because her force causes movement. Vaishali is not doing work because there is no movement in the direction of her force. So, the work done by Vaishali is zero.

Illustration of a hand pushing a book into the table while another hand is pushing a book across the table.

Figure 2: Pushing a book into the table

Can work done be negative?

Think about pushing a book across a table. The book moves forward because we apply a force. At the same time, the table exerts a friction force on the book in the opposite direction.

Remember that in formula for work (Equation 1), distance (d) is measured in the direction of the force. In this situation the book moves forward and the frictional force acts backward. So, the motion of the book is in the opposite direction to the friction force. When a force (like friction) tries to stop an object from moving forward, it is doing negative work because it is 'stealing' energy away from the motion.

If the force and the motion are in opposite directions, the work done is negative. Negative work does not mean "no work." It means that the force is acting against the motion. It tries to slow the object down or stop it.

QUESTION: We apply brakes to slow down and stop a cycle. The cycle is moving forward, but the braking force acts backward. Is the work done by the braking force positive or negative?

Energy

Energy is the capacity to do work. Let us understand this with an example. Imagine a stack of books kept on the floor. Let us pick up some of the books and place them on a table. To do this, we apply an upward force and move the books through a certain height. So, we have done work.

When we do work on the books, we are also transferring energy to them. The books on the floor have less energy than the books on the table that are at a higher position.

Similarly, imagine a football resting on the ground. It is not moving. When we kick the ball, we are applying a force to it over a short distance. So, we do work on the ball. As a result, energy is transferred from our body to the ball. This energy makes the ball move forward.

If we kick the ball harder, we are applying a larger force. So, we are doing more work, and more energy is transferred to the ball. The ball then moves faster and travels a longer distance. If the ball slows down and stops after some time, it is because forces like friction act on it and remove energy from it. In this way, by kicking the ball, we can clearly see how doing work transfers energy and causes motion.

Relationship between work and energy

Work and energy are closely related. When we do work on an object, we transfer energy to it. When an object does work on something else, it transfers energy from itself. Energy transferred to the object is positive work, and energy transferred from the object is negative work.

Work done on an object, its energy changes. We can say that work done on an object is the change in its energy.

$$\text{Work done on an object} = \text{change in the energy of the object.} \quad (3)$$

This means, if positive work is done, the energy of the object increases and if negative work is done, the energy of the object decreases. This idea is true even when the force is not constant or changes during motion.

The SI unit of both work and energy is the (J).

Our body gets energy to do work from the food we eat. The energy stored in food is often measured in calories (cal).

$$1 \text{ cal} = 4.2 \text{ J.} \quad (4)$$

EXERCISE: Look at the labels of packaged food items in your house. You may notice a table containing nutritional information. This table lists the energy contained in the food (often measured in kcal). Write down the energy (in calories) given on each label.

$$1 \text{ kcal} = 1000 \text{ cal}$$

Our body needs energy not just to run and play but to grow, think, and stay warm. Different foods give energy in different ways. Rice and idly give quick energy (carbohydrates), lentils, milk, and meat help build and repair the body (proteins), and butter, ghee or oil store energy for later (fats). A balanced diet gives the body the right mix of energy sources. Eating too little means not

enough energy for the day's work; eating far more than the body uses stores the extra energy as fat.

Different forms of Energy

Energy can exist in many different forms. We do not always see energy directly, but we can observe its effects.

For example, the books placed on a table have more energy than the books on the floor. This energy depends on their position. It is called **potential energy**.

Now imagine you push one of the books off the table. As the book falls, it starts moving. This energy due to motion is called **kinetic energy**. While the book is falling, its potential energy decreases and its kinetic energy increases. Just before it reaches the ground, most of its energy is in the form of kinetic energy. In this way, energy can change from potential energy to kinetic energy. Both potential energy and kinetic energy are forms of mechanical energy. **Mechanical energy** is the energy an object has due to its position or its motion.

In dam projects, water is stored at a height. This water has potential energy. When the water is released, it flows downward and gains kinetic energy. The moving water turns large turbines, which generate electricity. In this way, the energy of water is converted into electrical energy.

In an electric circuit, energy is stored in a battery. This is **electrical energy**. When the bulb glows, electrical energy is converted into **light energy** and some energy is also converted into heat.

Every electrical appliance has a power rating printed on it, in Watts (W). For example, an LED bulb might be 9 W while an oldstyle bulb is 60 W for similar brightness. The higher the Watt rating, the faster it uses energy. In India, appliances like fans, refrigerators, and ACs also carry a BEE star rating (1 to 5 stars). More stars means it does the same job using less energy. This is why replacing an old bulb with an LED, or choosing a 5-star fridge, saves electricity over time.

In your electricity bill, you are charged for the electrical energy you use. This is measured in kilowatt-hour (kW h).

$$1 \text{ kW h} = 3.6 \times 10^6 \text{ J} = 3600000 \text{ J} \quad (5)$$

EXAMPLE: When a 60 W bulb runs for 5 h (hour), it uses $60 \text{ W} \times 5 \text{ h} = 300 \text{ W h} = 0.3 \text{ kW h}$ of energy.

QUESTION: Check your home electricity bill. How many units (kW h) of energy have you used? Provide your answer in the unit

James Prescott Joule studied how mechanical energy and heat (thermal energy) are related and how one can change into the other. His work showed that different forms of energy are connected, helping scientists develop a unified understanding of energy.

He carried out careful experiments on heat and discovered how heat is produced when an electric current flows through a conductor. This relationship is known as Joule's law of heating.

Joule's work played an important role in the development of thermodynamics, the study of heat and energy.

In recognition of his contributions, he was elected a Fellow of the Royal Society in 1850. The SI unit of energy, the joule (J), is named in his honor.

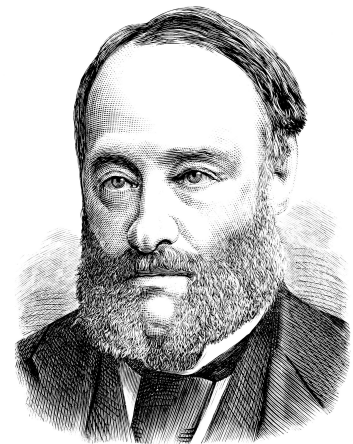


Figure 3: James Prescott Joule



Figure 4: Mettur Dam located across the river Kaveri in Salem.

of Joule.

Your electricity provider (like TANGEDCO in Tamil Nadu) charges you for every 'unit' (1 kW h) you use, at a certain rate per unit.

$$\text{Electricity bill amount} = \text{units used} \times \text{rate} \quad (6)$$

If a family uses 100 units in a month at INR 5 per unit, their bill is INR 500. Appliances with lower power ratings, used for less time, mean fewer units used and a smaller bill.

You may notice that when the electricity consumed at home is written in the unit of Joule, it is a large number. Measuring the electrical energy consumption in Joule is like measuring the distance between two cities in millimetre. Kilowatt-hour (kW h) is therefore a more convenient unit to measure electrical energy.

When we use an electric kettle electrical energy comes from the power supply. This energy passes through a metal coil (heating element) inside the kettle. The coil gets hot. This heat is transferred from the coil to the water. As a result, the temperature of the water increases and it eventually boils. The hot water has **thermal energy**.

When we beat a drum we use mechanical energy (movement of our hand). This energy is converted into **sound energy**. Similarly, speakers use electrical energy to produce sound.

The food we eat and fuels like petrol and gas store energy in the form of **chemical energy**. When we eat food, our body uses this energy to move, work, and stay warm. When fuel burns, chemical energy is released as heat and light.

QUESTION: A banana contains 100 calories of energy. How much energy is this in joules?

Energy is also stored inside the nuclei of atoms. This is called **nuclear energy**. It is released in special processes and is used, for example, to generate electricity in nuclear power plants. Nuclear energy powers the Sun.

Energy can change from one form to another. It does not disappear. When we observe any activity around us, we can often trace how energy changes form step by step.

For example, when you eat food and then run, the energy stored in the food (chemical energy) is used by your muscles to produce motion. Some of this energy also changes into heat, which keeps your body warm.

In another example, sunlight falls on wet clothes. The energy from sunlight is transferred to the water in the clothes. This energy makes the water evaporate, and the clothes dry.

Similarly, when you switch on a torch, the battery provides



Figure 5: Every 100 g of banana contains approximately 89 calories of energy.



Figure 6: Kudankulam Nuclear Power Plant in Tirunelveli

electrical energy. This energy is converted into light (and some heat), which helps us see in the dark.

In each of these cases, energy is not lost. It only changes from one form to another.

EXERCISE: Choose one of the examples above and draw arrows to show how energy changes form. Example: Food $\rightarrow$ muscle $\rightarrow$ motion $\rightarrow$ heat

QUESTION: When you ride a cycle, what forms of energy are involved?

QUESTION: When a fan is running, how does energy change form?

Law of Conservation of Energy

Energy can change from one form to another, but the total amount of energy remains the same. This idea is called the *Law of Conservation of Energy*.

Consider a roller coaster. When the coaster is at the top of a track, it has high potential energy because of its height. As it moves down, this potential energy changes into kinetic energy, and the coaster speeds up. When it climbs up again, kinetic energy changes back into potential energy, and the speed decreases.

In a similar way, think about riding a bicycle down a slope. At the top, you have more potential energy. As you move down, this changes into kinetic energy and you go faster. When you go up another slope, your speed decreases as kinetic energy changes back into potential energy.

In all these cases, energy is not lost. It only changes from one form to another. The total energy remains constant.

EXERCISE: Complete the table below. For each situation, describe what is happening and identify the form of energy before and after the change.

Energy from the Sun

The Sun is an important source of energy for life on Earth. It gives us light energy and heat energy.

We use sunlight in many ways. We use it for drying washed clothes. Plants use sunlight for photosynthesis to grow. Plants later becomes our food. Sunlight is a renewable source of energy.



Figure 7: Roller coaster in a theme park in Chennai.

What is happening?	Where is energy at the start?	Form of energy (before)	Form of energy (after)	Where does the energy go to?
We push a book and it moves	body	chemical energy	kinetic energy	book
...	...	...	...	...
...	...	...	...	...
...	...	...	...	...
...	...	...	...	...

This means it is naturally available and will not run out in a short time.

Diagram of working of a solar cooker

Sunlight, wind and flowing water are examples of renewable energy sources. These sources can be used again and again. Hydroelectric power and power generated from wind turbines are a renewable source of energy because it depends on the natural water cycle and flowing wind.



Figure 8: A solar cooker uses sunlight and can be used on a sunny day to cook food. A solar cooker does not use fuel like gas or coal. It does not produce smoke. It helps reduce energy waste.

Figure 9: The largest wind farm of India in Muppandal, Kanyakumari.



Coal, petrol, and natural gas are non-renewable energy sources. Non-renewable energy sources take millions of years to form and can get used up.

Energy from non-renewable sources is limited. If we waste it, it may not be available in the future. Using renewable sources and reducing unnecessary use helps save energy.

Vehicles like cars and bikes burn petrol (a non-renewable, chemical energy source) to move. Carrying one person by car uses

Figure 10: A farm in which uses Solar Power to run its water pumps and lights.

much more fuel per person than a bus or train carrying many people together. Walking or cycling for short distances uses no fuel at all. It uses the chemical energy from your food. Choosing a bus, cycle, or walking over a private vehicle when possible saves both fuel and money on tickets and petrol.”

QUESTION: Give one real-life situation where understanding energy can help reduce waste. Example: Switching off lights and fans when not in use to save electrical energy.

Image Source

Figure	Source	License
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Every 100 gof banana contains apporximately 89 calories of energy.	https://commons.wikimedia.org/wiki/File:Salem_Palaintains_of_Tamil_Nadu.jpg#/media/File:Salem_Palaintains_of_Tamil_Nadu.jpg	CC BY-SA 4.0
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